# SHARDA UNIVERSITY




# Student Project Diary
# Session : 2026-27
# Course Name : Capstone-1
# Course Code : CSP497

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## Title of The Project :
Intelligent DevOps Monitoring & Incident Response System


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# Project Diary (Student)

**Name:** Shruti Tyagi **Enrollment No. :** 2301010835

**Project Title:** Intelligent DevOps Monitoring & Incident Response System

| Week | Date of Visit | Status of Task assigned during last visit | Remarks | Task to be carried out till next visit | Signature of Guide |
|---|---|---|---|---|---|
| One | 22 Jul 2026 | First meeting; project group formed | | Finalise topic; study existing monitoring tools (Prometheus, Nagios, Datadog) | |
| Two | 29 Jul 2026 | Topic finalised; existing tools studied | | Literature survey: AIOps, anomaly detection, LLM incident analysis | |
| Three | 05 Aug 2026 | Literature survey done; gaps noted | | Write problem statement, objectives and scope | |
| Four | 12 Aug 2026 | Problem statement and objectives done | | Requirement analysis; select tools (Docker, Prometheus, Ollama) | |
| Five | 19 Aug 2026 | Requirements and tool list ready | | Design system architecture: monitor, detect, analyze, respond, verify | |
| Six | 26 Aug 2026 | Architecture and plan drafted | | Design IncidentEvent / ResponsePlan schemas, alert rules, safety policies | |
| Seven | 02 Sep 2026 | Schemas, rules and policies designed | | Set up Docker and Prometheus; base Flask app with / metrics | |

| Week | Date of Visit | Status of Task assigned during last visit | Remarks | Task to be carried out till next visit | Signature of Guide |
|---|---|---|---|---|---|
| Eight | 09 Sep 2026 | Environment set up; base app monitored by Prometheus | | Add fault injection: CPU, memory, latency, error faults | |
| Nine | 16 Sep 2026 | Fault injection working | | Draft Invention Disclosure Form (patent) | |
| Ten | 23 Sep 2026 | IDF first draft written | | System redesign: old design used only fixed runbooks; add LLM root-cause analysis, safety checks, re-analysis loop | |
| Eleven | 30 Sep 2026 | Redesign done: fixed-runbook design replaced by LLM analysis with safety checks and verification | | Multi-service app (frontend, API, Redis), alert rules, fault injector | |
| Twelve | 07 Oct 2026 | Done; also built event gateway, LLM analyzer, responder and verifier | | Web dashboard (Next.js), ML anomaly detector; finalise IDF | |
| Thirteen | 14 Oct 2026 | Dashboard, ML detector and final IDF done; 9 scenarios live | | Evaluation runner; stage 1 evaluation of 9 fault scenarios | |
| Fourteen | | | | | |
| Fifteen | | | | | |
| Sixteen | | | | | |

| Week | Date of Visit | Status of Task assigned during last visit | Remarks | Task to be carried out till next visit | Signature of Guide |
|---|---|---|---|---|---|
| Seventeen | | | | | |
| Eighteen | | | | | |
| Nineteen | | | | | |
| Twenty | | | | | |

**Name:** Shruti Tyagi **Enrollment No. :** 2301010835

**Project Title:** Intelligent DevOps Monitoring & Incident Response System

**Week No:** One **Date of Visit:** 22 Jul 2026

**Brief Description of task to be completed till next Visit:**

Finalise topic; study existing monitoring tools (Prometheus, Nagios, Datadog)

**Details of approaching the task and solving the task:**

Listed possible topics and chose intelligent DevOps monitoring, because handling an alert is still mostly manual work. Studied Prometheus (metrics and alert rules), Nagios (host and service checks) and Datadog (hosted monitoring). Noted that all three raise alerts but leave the diagnosis and the fix to an engineer.



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**Name:** Shruti Tyagi **Enrollment No. :** 2301010835

**Project Title:** Intelligent DevOps Monitoring & Incident Response System

**Week No:** Two **Date of Visit:** 29 Jul 2026

**Brief Description of task to be completed till next Visit:**

Literature survey: AIOps, anomaly detection, LLM incident analysis

**Details of approaching the task and solving the task:**

Read papers and articles on AIOps, anomaly detection (Isolation Forest, z-scores), root-cause analysis in microservices and LLM-based incident analysis. Recorded the gaps: most tools only detect problems, automated fixes have no safety check, and nothing confirms that a fix actually worked.



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**Name:** Shruti Tyagi **Enrollment No. :** 2301010835

**Project Title:** Intelligent DevOps Monitoring & Incident Response System

**Week No:** Three **Date of Visit:** 05 Aug 2026

**Brief Description of task to be completed till next Visit:**

Write problem statement, objectives and scope

**Details of approaching the task and solving the task:**

Wrote the problem statement: the time between an alert and its resolution is long because the analysis is manual. Set the objectives: detect faults, find the root-cause service, fix it safely and verify the fix. Fixed the scope to Docker Compose on a laptop with metrics only; Kubernetes is out of scope.



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**Project Title:** Intelligent DevOps Monitoring & Incident Response System

**Week No:** Four **Date of Visit:** 12 Aug 2026

**Brief Description of task to be completed till next Visit:**

Requirement analysis; select tools (Docker, Prometheus, Ollama)

**Details of approaching the task and solving the task:**

Listed the functional requirements (collect metrics, detect faults, analyze, respond, verify, show a dashboard) and the non-functional ones (runs on a laptop, safe, explainable). Selected Docker Compose, Prometheus with Alertmanager and cAdvisor, Python (Flask, FastAPI) and Ollama with a local Llama 3.1 8B model.



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**Project Title:** Intelligent DevOps Monitoring & Incident Response System

**Week No:** Five **Date of Visit:** 19 Aug 2026

**Brief Description of task to be completed till next Visit:**

Design system architecture: monitor, detect, analyze, respond, verify

**Details of approaching the task and solving the task:**

Drew the pipeline Monitor, Detect, Analyze, Decide, Respond and Verify. Placed an event gateway between the alerts and the analysis, and made the responder the only component allowed to change the system. Compared an ML-centric plan with an LLM-centric plan before choosing.



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**Name:** Shruti Tyagi **Enrollment No. :** 2301010835

**Project Title:** Intelligent DevOps Monitoring & Incident Response System

**Week No:** Six **Date of Visit:** 26 Aug 2026

**Brief Description of task to be completed till next Visit:**

Design IncidentEvent / ResponsePlan schemas, alert rules, safety policies

**Details of approaching the task and solving the task:**

Defined the IncidentEvent schema (alerts plus metric context) and the ResponsePlan schema (root cause, actions, confidence). Drafted alert rules for CPU, memory, latency, errors and service down. Wrote the safety policies: allowlisted actions, known targets, confidence of at least 0.7, a 5-minute cooldown, at most 3 actions per incident and approval for risky actions.



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**Project Title:** Intelligent DevOps Monitoring & Incident Response System

**Week No:** Seven **Date of Visit:** 02 Sep 2026

**Brief Description of task to be completed till next Visit:**

Set up Docker and Prometheus; base Flask app with /metrics

**Details of approaching the task and solving the task:**

Installed Docker Desktop and wrote a Compose file with Prometheus. Built a Flask test app that exposes /metrics with request counts and latency histograms, and checked in the Prometheus targets page that it was being scraped.



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**Project Title:** Intelligent DevOps Monitoring & Incident Response System

**Week No:** Eight **Date of Visit:** 09 Sep 2026

**Brief Description of task to be completed till next Visit:**

Add fault injection: CPU, memory, latency, error faults

**Details of approaching the task and solving the task:**

Added /inject and /reset endpoints for CPU, memory, latency and error faults; each fault clears itself after its duration. Made the memory fault write to its pages so that container memory really rises. Exported fault_*_active gauges as ground truth, used only for scoring.



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**Name:** Shruti Tyagi **Enrollment No. :** 2301010835

**Project Title:** Intelligent DevOps Monitoring & Incident Response System

**Week No:** Nine **Date of Visit:** 16 Sep 2026

**Brief Description of task to be completed till next Visit:**

Draft Invention Disclosure Form (patent)

**Details of approaching the task and solving the task:**

Wrote the Invention Disclosure Form: the problem, the prior art (monitoring systems, rule-based alerting, manual incident response), the proposed system and its novel parts. Prepared the block diagrams.



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**Name:** Shruti Tyagi **Enrollment No. :** 2301010835

**Project Title:** Intelligent DevOps Monitoring & Incident Response System

**Week No:** Ten **Date of Visit:** 23 Sep 2026

## Brief Description of task to be completed till next Visit:

System redesign: old design used only fixed runbooks; add LLM root-cause analysis, safety checks, re-analysis loop

## Details of approaching the task and solving the task:

Reviewed the first design: it relied on fixed runbooks and its LLM only wrote summaries, so unseen faults could not be handled and a failed fix was never re-analyzed. Redesigned it so the LLM finds the root cause and proposes actions, a policy validator checks every action before it runs, and a verifier re-checks and re-analyzes up to 3 times before escalating. Kept the runbooks as a fallback and moved from Kubernetes to Docker Compose.



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**Project Title:** Intelligent DevOps Monitoring & Incident Response System

**Week No:** Eleven **Date of Visit:** 30 Sep 2026

**Brief Description of task to be completed till next Visit:**

Multi-service app (frontend, API, Redis), alert rules, fault injector

**Details of approaching the task and solving the task:**

Split the app into frontend, api-service and Redis, with redis_exporter and cAdvisor. Wrote Prometheus alert rules with 30 s windows (about 40 s to detect) and an Alertmanager webhook. Built the fault injector with 9 scenarios that logs ground truth, plus Locust load. Also built the event gateway, the LLM analyzer with runbook fallback, and the responder with verifier, and ran the full loop on real containers.



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**Week No:** Twelve **Date of Visit:** 07 Oct 2026

**Brief Description of task to be completed till next Visit:**

Web dashboard (Next.js), ML anomaly detector; finalise IDF

**Details of approaching the task and solving the task:**

Built a Next.js dashboard with a service map, incident timelines, approvals and statistics. Recorded 78 minutes of normal traffic and trained the ML detector (robust z-scores plus Isolation Forest) to catch faults no rule covers, such as a slow memory leak. Finalised the IDF with claims and figures. Ran all 9 scenarios live: 9 of 9 detected.



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**Week No:** Thirteen **Date of Visit:** 14 Oct 2026

**Brief Description of task to be completed till next Visit:**

Evaluation runner; stage 1 evaluation of 9 fault scenarios

**Details of approaching the task and solving the task:**

Wrote an evaluation runner that injects each scenario, switches between the runbook-only baseline, the LLM baseline and the full system, and scores the results. Stage 1 (81 runs): 100% of faults detected and 0 unsafe actions; average detection time 27 s for the full system against 52 s for the runbook-only baseline.



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